

| Exp_ID    | Model            | Total_Params | Trainable_Param | Frozen_Params | Frozen_Layers | Unfrozen_Layers | IMG_Size | Batch_Size | Phase                    | LR       | Patience | Epochs_Run | Max_Epochs | Train_Acc | Val_Acc | Train_Loss | Val_Loss | Best_Epoch | Early_Stopped | Notes                                                       |                                   |  |  |
|-----------|------------------|--------------|-----------------|---------------|---------------|-----------------|----------|------------|--------------------------|----------|----------|------------|------------|-----------|---------|------------|----------|------------|---------------|-------------------------------------------------------------|-----------------------------------|--|--|
| EXP-01    | EfficientNetB0_R | 4,383,655    | 331,524         | 4,052,131     | 1             | 6               | 224      | 32         | Phase1 — Froze           | 0.001    | 5        | 5          | 6          | 20        | 26.41%  | 25.00%     | 1.5989   | 1.387      | 1             | TRUE                                                        | Black BG — 25% accuracy issue     |  |  |
| EXP-02    | EfficientNetB0_R | 4,383,655    | 331,524         | 4,052,131     | 1             | 6               | 224      | 32         | Phase2 — Fine T          | 1.00E-05 | 5        | 5          | 5          | 10        | 27.03%  | 25.00%     | 1.7092   | 1.3888     | 1             | TRUE                                                        | Fine tuning after Phase 1         |  |  |
| EXP-03    | RiceGrain_CNN    | 356,260      | 354,852         | 1,408         | 0             | 26              | 128      | 32         | Custom CNN —             | 0.001    | 8        | 9          | 40         | 38.67%    | 25.00%  | 1.2737     | 1.3958   | 1          | TRUE          | 3-black CNN on black BG preprocessed data                   |                                   |  |  |
| EXP-04    | VGG-16           | 138357544    | 14714688        | 123642856     | 24            | Block5(7)+FC(7) | 224      | 32         | 1.00E-04                 | -        | -        | 20         | 20         | 20        | 72%     | 53%        | 0.65     | 1.38       | 13            | FALSE                                                       | trained basic VGG-16              |  |  |
| EXP-05    | VGG-16(BN)       | 15114820     | 397572          | 14717248      | -             | -               | 224      | 32         | frozen base              | 1.00E-07 | 5        | 5          | 5          | 20        | 69.22%  | 55.31%     | 0.72     | 1.26       | 5             | TRUE                                                        | Used batch Normalisation          |  |  |
| EXP-06    | VGG-16-BN-FT     | 15,91M       | 7,48M           | 8,43M         | -             | -               | 224      | 32         | fine tuned last 4 layers | -        | 5        | 5          | 5          | 20        | 69.22%  | 55.31%     | 0.72     | 1.26       | 5             | TRUE                                                        | Fine Tuned last 4 layers          |  |  |
| EXP_07    | VGG16_Custom     | 134,276,932  | 126,641,668     | 7,635,264     | 36            | 6               | 224x224  | 32         | Transfer Learning        | 0.001    | 5        | 5          | 6          | 20        | 25.62%  | 25.00%     | 1.3865   | 1.3863     | 1             | TRUE                                                        | Transfer Learning - Adam lr=0.001 |  |  |
| EXP_08    | VGG16_Custom     | 134,276,932  | 126,641,668     | 7,635,264     | 36            | 6               | 224x224  | 32         | Transfer Learning        | 0.0001   | 5        | 12         | 20         | 99.69%    | 56.25%  | 0.0108     | 1.3323   | 7          | TRUE          | Transfer Learning - Adam lr=0.0001                          |                                   |  |  |
| EXP_09    | VGG16_Custom     | 134,276,932  | 126,641,668     | 7,635,264     | 36            | 6               | 224x224  | 32         | Transfer Learning        | 0.01     | 5        | 10         | 20         | 97.50%    | 55.00%  | 0.0835     | 1.4396   | 5          | TRUE          | Transfer Learning - SGD lr=0.01                             |                                   |  |  |
| EXP_10    | VGG16_Custom     | 134,276,932  | 126,641,668     | 7,635,264     | 36            | 6               | 224x224  | 32         | Augmented Tran           | 0.0001   | 5        | 16         | 20         | 82.58%    | 58.13%  | 0.4195     | 1.4813   | 11         | TRUE          | Augmented - Adam lr=0.0001                                  |                                   |  |  |
| EXP_11    | VGG16_Custom     | 134,276,932  | 126,641,668     | 7,635,264     | 36            | 6               | 224x224  | 32         | Augmented Tran           | 0.01     | 5        | 20         | 20         | 77.19%    | 59.38%  | 0.5595     | 1.2188   | 15         | FALSE         | Augmented - SGD lr=0.01                                     |                                   |  |  |
| EXP_12    | VGG16_Custom     | 134,276,932  | 126,641,668     | 7,635,264     | 36            | 6               | 224x224  | 32         | Augmented + Re           | 0.01     | 5        | 15         | 20         | 71.95%    | 52.81%  | 0.6841     | 1.2599   | 10         | TRUE          | SGD + Augmentation                                          |                                   |  |  |
| EXP_13    | VGG16_Custom     | 134,276,932  | 126,641,668     | 7,635,264     | 36            | 6               | 224x224  | 32         | Augmented + Re           | 0.01     | 5        | 10         | 20         | 67.42%    | 50.00%  | 0.7547     | 1.3271   | 5          | TRUE          | SGD + Augmentation + Weight Decay                           |                                   |  |  |
| EXP_14    | VGG16_Custom     | 134,276,932  | 126,641,668     | 7,635,264     | 36            | 6               | 224x224  | 32         | Augmented + Re           | 0.01     | 5        | 20         | 20         | 68.52%    | 50.00%  | 0.7034     | 1.3059   | 16         | FALSE         | SGD + Augmentation + Dropout 0.7                            |                                   |  |  |
| EXP_15    | VGG16_Custom     | 134,276,932  | 126,641,668     | 7,635,264     | 36            | 6               | 224x224  | 32         | Augmented + Re           | 0.01     | 5        | 20         | 20         | 71.41%    | 50.94%  | 0.6369     | 1.3142   | 19         | FALSE         | SGD + Aug + Dropout 0.7 + Weight Decay                      |                                   |  |  |
| EXP_16    | VGG16_Custom     | 134,276,932  | 126,641,668     | 7,635,264     | 36            | 6               | 224x224  | 32         | Transfer Learning        | 0.001    | 5        | 6          | 40         | 24.77%    | 25.00%  | 1.3869     | 1.3863   | 1          | TRUE          | Transfer Learning - Adam lr=0.001                           |                                   |  |  |
| EXP_17    | VGG16_Custom     | 134,276,932  | 126,641,668     | 7,635,264     | 36            | 6               | 224x224  | 32         | Transfer Learning        | 0.0001   | 5        | 11         | 40         | 100.00%   | 58.44%  | 0.0001     | 1.3182   | 6          | TRUE          | Transfer Learning - Adam lr=0.0001                          |                                   |  |  |
| EXP_18    | VGG16_Custom     | 134,276,932  | 126,641,668     | 7,635,264     | 36            | 6               | 224x224  | 32         | Transfer Learning        | 0.001    | 5        | 6          | 40         | 25.00%    | 25.00%  | 1.3867     | 1.3863   | 1          | TRUE          | Transfer Learning - Adam lr=0.001                           |                                   |  |  |
| EXP_19    | VGG16_Custom     | 134,276,932  | 126,641,668     | 7,635,264     | 36            | 6               | 224x224  | 32         | Transfer Learning        | 0.01     | 5        | 7          | 40         | 93.28%    | 52.19%  | 0.172      | 1.4838   | 2          | TRUE          | Transfer Learning - SGD lr=0.01                             |                                   |  |  |
| EXP_20    | VGG16_Custom     | 134,276,932  | 126,641,668     | 7,635,264     | 36            | 6               | 224x224  | 32         | Transfer Learning        | 0.0001   | 5        | 10         | 40         | 99.84%    | 54.37%  | 0.0088     | 1.5324   | 5          | TRUE          | Transfer Learning - Adam lr=0.0001                          |                                   |  |  |
| EXP_21    | VGG16_Custom     | 134,276,932  | 126,641,668     | 7,635,264     | 36            | 6               | 224x224  | 32         | Transfer Learning        | 0.01     | 5        | 10         | 40         | 95.78%    | 55.62%  | 0.1193     | 1.3045   | 5          | TRUE          | Transfer Learning - SGD lr=0.01                             |                                   |  |  |
| EXP_22    | VGG16_Custom     | 134,276,932  | 126,641,668     | 7,635,264     | 36            | 6               | 224x224  | 32         | Augmented Tran           | 0.0001   | 5        | 16         | 40         | 86.64%    | 58.75%  | 0.3647     | 1.3086   | 11         | TRUE          | Augmented - Adam lr=0.0001                                  |                                   |  |  |
| EXP_23    | VGG16_Custom     | 134,276,932  | 126,641,668     | 7,635,264     | 36            | 6               | 224x224  | 32         | Augmented Tran           | 0.01     | 5        | 16         | 40         | 74.06%    | 56.25%  | 0.6218     | 1.187    | 11         | TRUE          | Augmented - SGD lr=0.01                                     |                                   |  |  |
| EXP_25    | VGG16_Custom     | 134,276,932  | 126,641,668     | 7,635,264     | 36            | 6               | 224x224  | 32         | Augmented Tran           | 0.0001   | 5        | 20         | 40         | 86.48%    | 56.88%  | 0.3455     | 1.3996   | 15         | TRUE          | Augmented - Adam lr=0.0001                                  |                                   |  |  |
| EXP_26    | VGG16_Custom     | 134,276,932  | 126,641,668     | 7,635,264     | 36            | 6               | 224x224  | 32         | Augmented Tran           | 0.01     | 5        | 11         | 40         | 70.08%    | 48.75%  | 0.696      | 1.304    | 6          | TRUE          | Augmented - SGD lr=0.01                                     |                                   |  |  |
| EXP_27    | VGG16_Custom     | 134,276,932  | 126,641,668     | 7,635,264     | 36            | 6               | 224x224  | 32         | Augmented + Re           | 0.01     | 5        | 14         | 40         | 70.94%    | 51.25%  | 0.6596     | 1.2749   | 9          | TRUE          | SGD + Augmentation                                          |                                   |  |  |
| EXP_28    | VGG16_Custom     | 134,276,932  | 126,641,668     | 7,635,264     | 36            | 6               | 224x224  | 32         | Augmented + Re           | 0.01     | 5        | 10         | 40         | 66.41%    | 48.44%  | 0.7554     | 1.2174   | 5          | TRUE          | SGD + Augmentation + Weight Decay                           |                                   |  |  |
| EXP_29    | VGG16_Custom     | 134,276,932  | 126,641,668     | 7,635,264     | 36            | 6               | 224x224  | 32         | Augmented + Re           | 0.01     | 5        | 16         | 40         | 74.61%    | 53.75%  | 0.6104     | 1.2011   | 11         | TRUE          | SGD + Augmentation                                          |                                   |  |  |
| EXP_30    | VGG16_Custom     | 134,276,932  | 126,641,668     | 7,635,264     | 36            | 6               | 224x224  | 32         | Augmented + Re           | 0.01     | 5        | 20         | 40         | 69.69%    | 49.69%  | 0.6539     | 1.2083   | 15         | TRUE          | SGD + Augmentation + Dropout 0.7                            |                                   |  |  |
| EXP_31    | VGG16_Custom     | 134,276,932  | 126,641,668     | 7,635,264     | 36            | 6               | 224x224  | 32         | Augmented + Re           | 0.01     | 5        | 18         | 40         | 77.11%    | 52.50%  | 0.5317     | 1.2605   | 13         | TRUE          | SGD + Augmentation + Weight Decay                           |                                   |  |  |
| EXP_32    | VGG16_Custom     | 134,276,932  | 126,641,668     | 7,635,264     | 36            | 6               | 224x224  | 32         | Augmented + Re           | 0.01     | 5        | 12         | 40         | 66.80%    | 50.00%  | 0.7627     | 1.3622   | 7          | TRUE          | SGD + Aug + Dropout 0.7 + Weight Decay                      |                                   |  |  |
| EXP_33    | VGG16_Custom     | 134,276,932  | 126,641,668     | 7,635,264     | 36            | 6               | 224x224  | 32         | Augmented + Re           | 0.01     | 5        | 21         | 40         | 71.33%    | 53.12%  | 0.6592     | 1.3219   | 16         | TRUE          | SGD + Augmentation + Dropout 0.7                            |                                   |  |  |
| EXP_34    | VGG16_Custom     | 134,276,932  | 126,641,668     | 7,635,264     | 36            | 6               | 224x224  | 32         | Augmented + Re           | 0.01     | 5        | 13         | 40         | 63.67%    | 46.56%  | 0.801      | 1.2861   | 8          | TRUE          | SGD + Aug + Dropout 0.7 + Weight Decay                      |                                   |  |  |
| EFF_B0_35 | EfficientNet-B0  | 4,012,672    | 5,124           | 4,007,548     | 336           | 1               | 224x224  | 32         | transfer                 | 0.001    | 5        | 9          | 20         | 80.00%    | 80.00%  | 0.4907     | 0.5045   | 4          | TRUE          | Fold 1/5   Adam lr=0.001   EfficientNet-B0   transfer       |                                   |  |  |
| EFF_B0_36 | EfficientNet-B0  | 4,012,672    | 5,124           | 4,007,548     | 336           | 1               | 224x224  | 32         | transfer                 | 0.001    | 5        | 10         | 20         | 79.69%    | 80.31%  | 0.4991     | 0.5003   | 5          | TRUE          | Fold 2/5   Adam lr=0.001   EfficientNet-B0   transfer       |                                   |  |  |
| EFF_B0_37 | EfficientNet-B0  | 4,012,672    | 5,124           | 4,007,548     | 336           | 1               | 224x224  | 32         | transfer                 | 0.001    | 5        | 9          | 20         | 80.39%    | 80.31%  | 0.4901     | 0.5018   | 4          | TRUE          | Fold 3/5   Adam lr=0.001   EfficientNet-B0   transfer       |                                   |  |  |
| EFF_B0_38 | EfficientNet-B0  | 4,012,672    | 5,124           | 4,007,548     | 336           | 1               | 224x224  | 32         | transfer                 | 0.001    | 5        | 7          | 20         | 79.61%    | 80.62%  | 0.4921     | 0.5241   | 2          | TRUE          | Fold 4/5   Adam lr=0.001   EfficientNet-B0   transfer       |                                   |  |  |
| EFF_B0_39 | EfficientNet-B0  | 4,012,672    | 5,124           | 4,007,548     | 336           | 1               | 224x224  | 32         | transfer                 | 0.001    | 5        | 9          | 20         | 79.92%    | 81.25%  | 0.4893     | 0.4846   | 4          | TRUE          | Fold 1/5   Adam lr=0.001   EfficientNet-B0   transfer       |                                   |  |  |
| EFF_B0_40 | EfficientNet-B0  | 4,012,672    | 5,124           | 4,007,548     | 336           | 1               | 224x224  | 32         | transfer                 | 0.001    | 5        | 10         | 20         | 80.16%    | 80.00%  | 0.4895     | 0.4932   | 5          | TRUE          | Fold 2/5   Adam lr=0.001   EfficientNet-B0   transfer       |                                   |  |  |
| EFF_B0_41 | EfficientNet-B0  | 4,012,672    | 5,124           | 4,007,548     | 336           | 1               | 224x224  | 32         | transfer                 | 0.001    | 5        | 7          | 20         | 79.77%    | 80.00%  | 0.4896     | 0.5036   | 2          | TRUE          | Fold 3/5   Adam lr=0.001   EfficientNet-B0   transfer       |                                   |  |  |
| EFF_B0_42 | EfficientNet-B0  | 4,012,672    | 5,124           | 4,007,548     | 336           | 1               | 224x224  | 32         | transfer                 | 0.001    | 5        | 8          | 20         | 80.23%    | 80.62%  | 0.4919     | 0.4992   | 3          | TRUE          | Fold 1/5   Adam lr=0.001   EfficientNet-B0   transfer       |                                   |  |  |
| EFF_B0_43 | EfficientNet-B0  | 4,012,672    | 5,124           | 4,007,548     | 336           | 1               | 224x224  | 32         | transfer                 | 0.001    | 5        | 7          | 20         | 79.77%    | 80.31%  | 0.5004     | 0.5314   | 2          | TRUE          | Fold 4/5   Adam lr=0.001   EfficientNet-B0   transfer       |                                   |  |  |
| EFF_B0_44 | EfficientNet-B0  | 4,012,672    | 5,124           | 4,007,548     | 336           | 1               | 224x224  | 32         | transfer                 | 0.001    | 5        | 8          | 20         | 79.84%    | 81.25%  | 0.4917     | 0.4903   | 3          | TRUE          | Fold 1/5   Adam lr=0.001   EfficientNet-B0   transfer       |                                   |  |  |
| EFF_B0_45 | EfficientNet-B0  | 4,012,672    | 5,124           | 4,007,548     | 336           | 1               | 224x224  | 32         | transfer                 | 0.001    | 5        | 7          | 20         | 79.84%    | 79.69%  | 0.4949     | 0.4983   | 2          | TRUE          | Fold 2/5   Adam lr=0.001   EfficientNet-B0   transfer       |                                   |  |  |
| EFF_B0_46 | EfficientNet-B0  | 4,012,672    | 5,124           | 4,007,548     | 336           | 1               | 224x224  | 32         | transfer                 | 0.001    | 5        | 8          | 20         | 79.45%    | 80.62%  | 0.4978     | 0.5072   | 3          | TRUE          | Fold 3/5   Adam lr=0.001   EfficientNet-B0   transfer       |                                   |  |  |
| EFF_B0_47 | EfficientNet-B0  | 4,012,672    | 5,124           | 4,007,548     | 336           | 1               | 224x224  | 32         | transfer                 | 0.001    | 5        | 8          | 20         | 80.23%    | 80.31%  | 0.4908     | 0.5199   | 3          | TRUE          | Fold 4/5   Adam lr=0.001   EfficientNet-B0   transfer       |                                   |  |  |
| EFF_B0_48 | EfficientNet-B0  | 4,012,672    | 5,124           | 4,007,548     | 336           | 1               | 224x224  | 32         | transfer                 | 0.001    | 5        | 6          | 20         | 80.39%    | 78.75%  | 0.4988     | 0.5111   | 1          | TRUE          | Fold 5/5   Adam lr=0.001   EfficientNet-B0   transfer       |                                   |  |  |
| EFF_B0_49 | EfficientNet-B0  | 4,012,672    | 1,134,516       | 2,878,156     | 326           | 11              | 224x224  | 32         | finetune_last2           | 0.001    | 5        | 9          | 20         | 81.17%    | 82.19%  | 0.4273     | 0.4544   | 4          | TRUE          | Fold 1/5   Adam lr=0.001   EfficientNet-B0   finetune_last2 |                                   |  |  |
| EFF_B0_50 | EfficientNet-B0  | 4,012,672    | 1,134,516       | 2,878,156     | 326           | 11              | 224x224  | 32         | finetune_last2           | 0.001    | 5        | 6          | 20         | 80.94%    | 80.00%  | 0.4459     | 0.4692   | 1          | TRUE          | Fold 2/5   Adam lr=0.001   EfficientNet-B0   finetune_last2 |                                   |  |  |
| EFF_B0_51 | EfficientNet-B0  | 4,012,672    | 1,134,516       | 2,878,156     | 326           | 11              | 224x224  | 32         | finetune_last2           | 0.001    | 5        | 6          | 20         | 80.70%    | 80.94%  | 0.4285     | 0.4908   | 1          | TRUE          | Fold 3/5   Adam lr=0.001   EfficientNet-B0   finetune_last2 |                                   |  |  |
| EFF_B0_52 | EfficientNet-B0  | 4,012,672    | 1,134,516       | 2,878,156     | 326           | 11              | 224x224  | 32         | finetune_last2           | 0.001    | 5        | 10         | 20         | 82.11%    | 80.31%  | 0.412      | 0.5041   | 5          | TRUE          | Fold 4/5   Adam lr=0.001   EfficientNet-B0   finetune_last2 |                                   |  |  |
| EFF_B0_53 | EfficientNet-B0  | 4,012,672    | 1,134,516       | 2,878,156     | 326           | 11              | 224x224  | 32         | finetune_last2           | 0.001    | 5        | 12         | 20         | 83.12%    | 81.56%  | 0.3971     | 0.4611   | 7          | TRUE          | Fold 5/5   Adam lr=0.001   EfficientNet-B0   finetune_last2 |                                   |  |  |
| EFF_B0_54 | EfficientNet-B0  | 4,012,672    | 4,012,672       | 0             | 206           | 131             | 224x224  | 32         | full                     | 0.001    | 5        | 10         | 20         | 80.62%    | 82.81%  | 0.4574     | 0.4342   | 5          | TRUE          | Fold 1/5   Adam lr=0.001   EfficientNet-B0   full           |                                   |  |  |
| EFF_B0_55 | EfficientNet-B0  | 4,012,672    | 5,124           | 4,007,548     | 336           | 1               | 224x224  | 32         | transfer                 | 0.001    | 5        | 6          | 20         | 79.84%    | 79.69%  | 0.4979     | 0.5112   | 1          | TRUE          | Fold 5/5   Adam lr=0.001   EfficientNet-B0   transfer       |                                   |  |  |
| EFF_B0_56 | EfficientNet-B0  | 4,012,672    | 4,012,672       | 0             | 206           | 131             | 224x224  | 32         | full                     | 0.001    | 5        | 9          | 20         | 81.09%    | 80.00%  | 0.4308     | 0.4958   | 4          | TRUE          | Fold 2/5   Adam lr=0.001   EfficientNet-B0   full           |                                   |  |  |
| EFF_B0_57 | EfficientNet-B0  | 4,012,672    | 4,012,672       | 0             | 206           | 131             | 224x224  | 32         | full                     | 0.001    | 5        | 15         | 20         | 82.81%    | 83.12%  | 0.407      |          |            |               |                                                             |                                   |  |  |

|                                                                                 |                 |            |            |            |     |     |         |  |    |                |       |   |    |    |        |        |        |        |    |      |                                                             |
|---------------------------------------------------------------------------------|-----------------|------------|------------|------------|-----|-----|---------|--|----|----------------|-------|---|----|----|--------|--------|--------|--------|----|------|-------------------------------------------------------------|
| EFF_B2_76                                                                       | EfficientNet-B2 | 7,706,630  | 5,636      | 7,700,994  | 472 | 1   | 224x224 |  | 32 | transfer       | 0.001 | 5 | 8  | 20 | 79.69% | 80.31% | 0.4787 | 0.5086 | 3  | TRUE | Fold 1/5   Adam lr=0.001   EfficientNet-B2   transfer       |
| EFF_B2_77                                                                       | EfficientNet-B2 | 7,706,630  | 5,636      | 7,700,994  | 472 | 1   | 224x224 |  | 32 | transfer       | 0.001 | 5 | 7  | 20 | 79.53% | 80.31% | 0.4986 | 0.4927 | 2  | TRUE | Fold 2/5   Adam lr=0.001   EfficientNet-B2   transfer       |
| EFF_B2_78                                                                       | EfficientNet-B2 | 7,706,630  | 5,636      | 7,700,994  | 472 | 1   | 224x224 |  | 32 | transfer       | 0.001 | 5 | 7  | 20 | 79.92% | 79.69% | 0.4904 | 0.5209 | 2  | TRUE | Fold 3/5   Adam lr=0.001   EfficientNet-B2   transfer       |
| EFF_B2_79                                                                       | EfficientNet-B2 | 7,706,630  | 5,636      | 7,700,994  | 472 | 1   | 224x224 |  | 32 | transfer       | 0.001 | 5 | 7  | 20 | 79.92% | 79.38% | 0.4881 | 0.5096 | 2  | TRUE | Fold 4/5   Adam lr=0.001   EfficientNet-B2   transfer       |
| EFF_B2_80                                                                       | EfficientNet-B2 | 7,706,630  | 5,636      | 7,700,994  | 472 | 1   | 224x224 |  | 32 | transfer       | 0.001 | 5 | 7  | 20 | 80.08% | 79.38% | 0.4995 | 0.5279 | 2  | TRUE | Fold 5/5   Adam lr=0.001   EfficientNet-B2   transfer       |
| EFF_B2_81                                                                       | EfficientNet-B2 | 7,706,630  | 3,239,888  | 4,466,742  | 454 | 19  | 224x224 |  | 32 | finetune_last2 | 0.001 | 5 | 13 | 20 | 82.81% | 82.19% | 0.3777 | 0.4671 | 8  | TRUE | Fold 1/5   Adam lr=0.001   EfficientNet-B2   finetune_last2 |
| EFF_B0_82                                                                       | EfficientNet-B0 | 4,012,672  | 1,134,516  | 2,878,156  | 326 | 11  | 224x224 |  | 32 | finetune_last2 | 0.001 | 5 | 8  | 20 | 81.56% | 80.62% | 0.4408 | 0.4727 | 3  | TRUE | Fold 2/5   Adam lr=0.001   EfficientNet-B0   finetune_last2 |
| EFF_B2_83                                                                       | EfficientNet-B2 | 7,706,630  | 3,239,888  | 4,466,742  | 454 | 19  | 224x224 |  | 32 | finetune_last2 | 0.001 | 5 | 9  | 20 | 82.11% | 81.25% | 0.402  | 0.4557 | 4  | TRUE | Fold 2/5   Adam lr=0.001   EfficientNet-B2   finetune_last2 |
| EFF_B2_84                                                                       | EfficientNet-B2 | 7,706,630  | 3,239,888  | 4,466,742  | 454 | 19  | 224x224 |  | 32 | finetune_last2 | 0.001 | 5 | 14 | 20 | 83.75% | 82.19% | 0.3729 | 0.4751 | 9  | TRUE | Fold 3/5   Adam lr=0.001   EfficientNet-B2   finetune_last2 |
| EFF_B2_85                                                                       | EfficientNet-B2 | 7,706,630  | 3,239,888  | 4,466,742  | 454 | 19  | 224x224 |  | 32 | finetune_last2 | 0.001 | 5 | 11 | 20 | 82.66% | 81.25% | 0.4009 | 0.4605 | 6  | TRUE | Fold 4/5   Adam lr=0.001   EfficientNet-B2   finetune_last2 |
| EFF_B2_86                                                                       | EfficientNet-B2 | 7,706,630  | 3,239,888  | 4,466,742  | 454 | 19  | 224x224 |  | 32 | finetune_last2 | 0.001 | 5 | 14 | 20 | 83.91% | 81.25% | 0.3551 | 0.4803 | 9  | TRUE | Fold 5/5   Adam lr=0.001   EfficientNet-B2   finetune_last2 |
| EFF_B2_87                                                                       | EfficientNet-B2 | 7,706,630  | 7,706,630  | 0          | 288 | 185 | 224x224 |  | 32 | full           | 0.001 | 5 | 14 | 20 | 81.41% | 82.81% | 0.431  | 0.4143 | 9  | TRUE | Fold 1/5   Adam lr=0.001   EfficientNet-B2   full           |
| EFF_B2_88                                                                       | EfficientNet-B2 | 7,706,630  | 7,706,630  | 0          | 288 | 185 | 224x224 |  | 32 | full           | 0.001 | 5 | 8  | 20 | 80.94% | 80.94% | 0.4493 | 0.4624 | 3  | TRUE | Fold 2/5   Adam lr=0.001   EfficientNet-B2   full           |
| EFF_B2_89                                                                       | EfficientNet-B2 | 7,706,630  | 7,706,630  | 0          | 288 | 185 | 224x224 |  | 32 | full           | 0.001 | 5 | 19 | 20 | 84.38% | 81.56% | 0.3445 | 0.4431 | 14 | TRUE | Fold 3/5   Adam lr=0.001   EfficientNet-B2   full           |
| EFF_B2_90                                                                       | EfficientNet-B2 | 7,706,630  | 7,706,630  | 0          | 288 | 185 | 224x224 |  | 32 | full           | 0.001 | 5 | 18 | 20 | 83.98% | 82.19% | 0.3755 | 0.4421 | 13 | TRUE | Fold 4/5   Adam lr=0.001   EfficientNet-B2   full           |
| EFF_B2_91                                                                       | EfficientNet-B2 | 7,706,630  | 7,706,630  | 0          | 288 | 185 | 224x224 |  | 32 | full           | 0.001 | 5 | 7  | 20 | 81.17% | 80.00% | 0.4323 | 0.4535 | 2  | TRUE | Fold 5/5   Adam lr=0.001   EfficientNet-B2   full           |
| EFF_B3_92                                                                       | EfficientNet-B3 | 10,702,380 | 6,148      | 10,696,232 | 532 | 1   | 224x224 |  | 32 | transfer       | 0.001 | 5 | 13 | 20 | 80.00% | 80.31% | 0.4899 | 0.4864 | 8  | TRUE | Fold 1/5   Adam lr=0.001   EfficientNet-B3   transfer       |
| EFF_B3_93                                                                       | EfficientNet-B3 | 10,702,380 | 6,148      | 10,696,232 | 532 | 1   | 224x224 |  | 32 | transfer       | 0.001 | 5 | 7  | 20 | 80.16% | 80.31% | 0.4954 | 0.5087 | 2  | TRUE | Fold 2/5   Adam lr=0.001   EfficientNet-B3   transfer       |
| EFF_B0_94                                                                       | EfficientNet-B0 | 4,012,672  | 1,134,516  | 2,878,156  | 326 | 11  | 224x224 |  | 32 | finetune_last2 | 0.001 | 5 | 11 | 20 | 82.58% | 81.88% | 0.4048 | 0.4867 | 6  | TRUE | Fold 3/5   Adam lr=0.001   EfficientNet-B0   finetune_last2 |
| EFF_B3_95                                                                       | EfficientNet-B3 | 10,702,380 | 6,148      | 10,696,232 | 532 | 1   | 224x224 |  | 32 | transfer       | 0.001 | 5 | 6  | 20 | 79.77% | 78.75% | 0.5023 | 0.5202 | 1  | TRUE | Fold 3/5   Adam lr=0.001   EfficientNet-B3   transfer       |
| EFF_B3_96                                                                       | EfficientNet-B3 | 10,702,380 | 6,148      | 10,696,232 | 532 | 1   | 224x224 |  | 32 | transfer       | 0.001 | 5 | 16 | 20 | 80.23% | 81.25% | 0.4709 | 0.4976 | 11 | TRUE | Fold 4/5   Adam lr=0.001   EfficientNet-B3   transfer       |
| EFF_B3_97                                                                       | EfficientNet-B3 | 10,702,380 | 6,148      | 10,696,232 | 532 | 1   | 224x224 |  | 32 | transfer       | 0.001 | 5 | 7  | 20 | 79.84% | 79.38% | 0.4987 | 0.5218 | 2  | TRUE | Fold 5/5   Adam lr=0.001   EfficientNet-B3   transfer       |
| EFF_B3_98                                                                       | EfficientNet-B3 | 10,702,380 | 3,883,262  | 6,819,118  | 514 | 19  | 224x224 |  | 32 | finetune_last2 | 0.001 | 5 | 8  | 20 | 81.56% | 81.56% | 0.4297 | 0.4479 | 3  | TRUE | Fold 1/5   Adam lr=0.001   EfficientNet-B3   finetune_last2 |
| EFF_B3_99                                                                       | EfficientNet-B3 | 10,702,380 | 3,883,262  | 6,819,118  | 514 | 19  | 224x224 |  | 32 | finetune_last2 | 0.001 | 5 | 9  | 20 | 81.56% | 80.94% | 0.4261 | 0.4554 | 4  | TRUE | Fold 2/5   Adam lr=0.001   EfficientNet-B3   finetune_last2 |
| EFF_B3_100                                                                      | EfficientNet-B3 | 10,702,380 | 3,883,262  | 6,819,118  | 514 | 19  | 224x224 |  | 32 | finetune_last2 | 0.001 | 5 | 6  | 20 | 81.33% | 80.00% | 0.4416 | 0.4671 | 1  | TRUE | Fold 3/5   Adam lr=0.001   EfficientNet-B3   finetune_last2 |
| EFF_B3_101                                                                      | EfficientNet-B3 | 10,702,380 | 3,883,262  | 6,819,118  | 514 | 19  | 224x224 |  | 32 | finetune_last2 | 0.001 | 5 | 7  | 20 | 81.56% | 80.62% | 0.4252 | 0.4815 | 2  | TRUE | Fold 4/5   Adam lr=0.001   EfficientNet-B3   finetune_last2 |
| EFF_B3_102                                                                      | EfficientNet-B3 | 10,702,380 | 3,883,262  | 6,819,118  | 514 | 19  | 224x224 |  | 32 | finetune_last2 | 0.001 | 5 | 8  | 20 | 81.09% | 80.62% | 0.4418 | 0.4621 | 3  | TRUE | Fold 5/5   Adam lr=0.001   EfficientNet-B3   finetune_last2 |
| EFF_B3_103                                                                      | EfficientNet-B3 | 10,702,380 | 10,702,380 | 0          | 324 | 209 | 224x224 |  | 32 | full           | 0.001 | 5 | 7  | 20 | 80.94% | 81.56% | 0.4298 | 0.4282 | 2  | TRUE | Fold 1/5   Adam lr=0.001   EfficientNet-B3   full           |
| EFF_B3_104                                                                      | EfficientNet-B3 | 10,702,380 | 10,702,380 | 0          | 324 | 209 | 224x224 |  | 32 | full           | 0.001 | 5 | 6  | 20 | 80.94% | 80.31% | 0.4519 | 0.4695 | 1  | TRUE | Fold 2/5   Adam lr=0.001   EfficientNet-B3   full           |
| EFF_B3_105                                                                      | EfficientNet-B3 | 10,702,380 | 10,702,380 | 0          | 324 | 209 | 224x224 |  | 32 | full           | 0.001 | 5 | 6  | 20 | 81.48% | 80.00% | 0.439  | 0.478  | 1  | TRUE | Fold 3/5   Adam lr=0.001   EfficientNet-B3   full           |
| EFF_B3_106                                                                      | EfficientNet-B3 | 10,702,380 | 10,702,380 | 0          | 324 | 209 | 224x224 |  | 32 | full           | 0.001 | 5 | 12 | 20 | 83.36% | 81.25% | 0.4023 | 0.4654 | 7  | TRUE | Fold 4/5   Adam lr=0.001   EfficientNet-B3   full           |
| EFF_B0_107                                                                      | EfficientNet-B0 | 4,012,672  | 1,134,516  | 2,878,156  | 326 | 11  | 224x224 |  | 32 | finetune_last2 | 0.001 | 5 | 9  | 20 | 82.27% | 81.56% | 0.4201 | 0.4527 | 4  | TRUE | Fold 4/5   Adam lr=0.001   EfficientNet-B0   finetune_last2 |
| EFF_B3_108                                                                      | EfficientNet-B3 | 10,702,380 | 10,702,380 | 0          | 324 | 209 | 224x224 |  | 32 | full           | 0.001 | 5 | 11 | 20 | 82.58% | 82.19% | 0.4094 | 0.4101 | 6  | TRUE | Fold 5/5   Adam lr=0.001   EfficientNet-B3   full           |
| EFF_B4_109                                                                      | EfficientNet-B4 | 17,555,788 | 7,172      | 17,548,616 | 652 | 1   | 224x224 |  | 32 | transfer       | 0.001 | 5 | 6  | 20 | 79.69% | 79.38% | 0.5269 | 0.5465 | 1  | TRUE | Fold 1/5   Adam lr=0.001   EfficientNet-B4   transfer       |
| EFF_B4_110                                                                      | EfficientNet-B4 | 17,555,788 | 7,172      | 17,548,616 | 652 | 1   | 224x224 |  | 32 | transfer       | 0.001 | 5 | 6  | 20 | 79.69% | 79.38% | 0.5165 | 0.5541 | 1  | TRUE | Fold 2/5   Adam lr=0.001   EfficientNet-B4   transfer       |
| EFF_B4_111                                                                      | EfficientNet-B4 | 17,555,788 | 7,172      | 17,548,616 | 652 | 1   | 224x224 |  | 32 | transfer       | 0.001 | 5 | 10 | 20 | 80.08% | 78.75% | 0.4868 | 0.5509 | 5  | TRUE | Fold 3/5   Adam lr=0.001   EfficientNet-B4   transfer       |
| EFF_B4_112                                                                      | EfficientNet-B4 | 17,555,788 | 7,172      | 17,548,616 | 652 | 1   | 224x224 |  | 32 | transfer       | 0.001 | 5 | 15 | 20 | 79.92% | 80.31% | 0.4823 | 0.5061 | 10 | TRUE | Fold 4/5   Adam lr=0.001   EfficientNet-B4   transfer       |
| EFF_B4_113                                                                      | EfficientNet-B4 | 17,555,788 | 7,172      | 17,548,616 | 652 | 1   | 224x224 |  | 32 | transfer       | 0.001 | 5 | 13 | 20 | 79.84% | 80.62% | 0.4995 | 0.4978 | 8  | TRUE | Fold 5/5   Adam lr=0.001   EfficientNet-B4   transfer       |
| EFF_B4_114                                                                      | EfficientNet-B4 | 17,555,788 | 5,283,576  | 12,272,212 | 634 | 19  | 224x224 |  | 32 | finetune_last2 | 0.001 | 5 | 13 | 20 | 82.03% | 81.88% | 0.4242 | 0.433  | 8  | TRUE | Fold 1/5   Adam lr=0.001   EfficientNet-B4   finetune_last2 |
| EFF_B4_115                                                                      | EfficientNet-B4 | 17,555,788 | 5,283,576  | 12,272,212 | 634 | 19  | 224x224 |  | 32 | finetune_last2 | 0.001 | 5 | 9  | 20 | 81.72% | 80.31% | 0.4298 | 0.4509 | 4  | TRUE | Fold 2/5   Adam lr=0.001   EfficientNet-B4   finetune_last2 |
| EFF_B4_116                                                                      | EfficientNet-B4 | 17,555,788 | 5,283,576  | 12,272,212 | 634 | 19  | 224x224 |  | 32 | finetune_last2 | 0.001 | 5 | 6  | 20 | 80.47% | 80.00% | 0.4482 | 0.4699 | 1  | TRUE | Fold 3/5   Adam lr=0.001   EfficientNet-B4   finetune_last2 |
| EFF_B4_117                                                                      | EfficientNet-B4 | 17,555,788 | 5,283,576  | 12,272,212 | 634 | 19  | 224x224 |  | 32 | finetune_last2 | 0.001 | 5 | 12 | 20 | 82.11% | 81.88% | 0.4167 | 0.454  | 7  | TRUE | Fold 4/5   Adam lr=0.001   EfficientNet-B4   finetune_last2 |
| EFF_B4_118                                                                      | EfficientNet-B4 | 17,555,788 | 5,283,576  | 12,272,212 | 634 | 19  | 224x224 |  | 32 | finetune_last2 | 0.001 | 5 | 11 | 20 | 81.25% | 80.31% | 0.4236 | 0.4667 | 6  | TRUE | Fold 5/5   Adam lr=0.001   EfficientNet-B4   finetune_last2 |
| EFF_B4_full_EXI2026-06-10 09:2' preprocessed_v2 Resize 224x224, EfficientNet-B4 |                 | 17,555,788 | 17,555,788 | 0          | 396 | 257 | 224x224 |  | 32 | full           | 0.001 | 5 | 8  | 20 | 82.73% | 80.94% | 0.4008 | 0.4473 | 3  | TRUE | Fold 1/5   Adam lr=0.001   EfficientNet-B4   full           |
| EFF_B0_fineturn2026-06-10 09:2' preprocessed_v2 Resize 224x224, EfficientNet-B0 |                 | 4,012,672  | 1,134,516  | 2,878,156  | 326 | 11  | 224x224 |  | 32 | finetune_last2 | 0.001 | 5 | 11 | 20 | 82.03% | 79.06% | 0.4154 | 0.4782 | 6  | TRUE | Fold 5/5   Adam lr=0.001   EfficientNet-B0   finetune_last2 |
| EFF_B4_full_EXI2026-06-10 09:2' preprocessed_v2 Resize 224x224, EfficientNet-B4 |                 | 17,555,788 | 17,555,788 | 0          | 396 | 257 | 224x224 |  | 32 | full           | 0.001 | 5 | 9  | 20 | 84.53% | 82.19% | 0.3684 | 0.4322 | 4  | TRUE | Fold 2/5   Adam lr=0.001   EfficientNet-B4   full           |
| EFF_B4_full_EXI2026-06-10 09:2' preprocessed_v2 Resize 224x224, EfficientNet-B4 |                 | 17,555,788 | 17,555,788 | 0          | 396 | 257 | 224x224 |  | 32 | full           | 0.001 | 5 | 8  | 20 | 84.77% | 81.88% | 0.3615 | 0.4587 | 3  | TRUE | Fold 3/5   Adam lr=0.001   EfficientNet-B4   full           |
| EFF_B4_full_EXI2026-06-10 09:3' preprocessed_v2 Resize 224x224, EfficientNet-B4 |                 | 17,555,788 | 17,555,788 | 0          | 396 | 257 | 224x224 |  | 32 | full           | 0.001 | 5 | 14 | 20 | 86.88% | 83.12% | 0.2973 | 0.4547 | 9  | TRUE | Fold 4/5   Adam lr=0.001   EfficientNet-B4   full           |
| EFF_B4_full_EXI2026-06-10 09:3' preprocessed_v2 Resize 224x224, EfficientNet-B4 |                 | 17,555,788 | 17,555,788 | 0          | 396 | 257 | 224x224 |  | 32 | full           | 0.001 | 5 | 13 | 20 | 87.34% | 82.81% | 0.3077 | 0.4461 | 8  | TRUE | Fold 5/5   Adam lr=0.001   EfficientNet-B4   full           |
| EFF_B0_transfer2026-06-10 09:5' preprocessed_v2 Resize 224x224, EfficientNet-B0 |                 | 4,012,672  | 5,124      | 4,007,548  | 336 | 1   | 224x224 |  | 32 | transfer       | 0.001 | 5 | 8  | 20 | 79.84% | 81.25% | 0.4917 | 0.4903 | 3  | TRUE | Fold 1/5   Adam lr=0.001   EfficientNet-B0   transfer       |
| EFF_B0_transfer2026-06-10 09:5' preprocessed_v2 Resize 224x224, EfficientNet-B0 |                 | 4,012,672  | 5,124      | 4,007,548  | 336 | 1   | 224x224 |  | 32 | transfer       | 0.001 | 5 | 7  | 20 | 79.84% | 79.69% | 0.4949 | 0.4983 | 2  | TRUE | Fold 2/5   Adam lr=0.001   EfficientNet-B0   transfer       |
| EFF_B0_transfer2026-06-10 09:5' preprocessed_v2 Resize 224x224, EfficientNet-B0 |                 | 4,012,672  | 5,124      | 4,007,548  | 336 | 1   | 224x224 |  | 32 | transfer       | 0.001 | 5 | 8  | 20 | 79.45% | 80.62% | 0.4978 | 0.5072 | 3  | TRUE | Fold 3/5   Adam lr=0.001   EfficientNet-B0   transfer       |
| EFF_B0_transfer2026-06-10 09:5' preprocessed_v2 Resize 224x224, EfficientNet-B0 |                 | 4,012,672  | 5,124      | 4,007,548  | 336 | 1   | 224x224 |  | 32 | transfer       | 0.001 | 5 | 8  | 20 | 80.23% | 80.31% | 0.4908 | 0.5199 | 3  | TRUE | Fold 4/5   Adam lr=0.001   EfficientNet-B0   transfer       |
| EFF_B0_transfer2026-06-10 09:5' preprocessed_v2 Resize 224x224, EfficientNet-B0 |                 | 4,012,672  | 5,124      | 4,007,548  | 336 | 1   | 224x224 |  | 32 | transfer       | 0.001 | 5 | 6  | 20 | 80.39% | 78.75% | 0.4988 | 0.5111 | 1  | TRUE | Fold 5/5   Adam lr=0.001   EfficientNet-B0   transfer       |
| EFF_B0_fineturn2026-06-10 09:5' preprocessed_v2 Resize 224x224, EfficientNet-B0 |                 | 4,012,672  | 1,134,516  | 2,878      |     |     |         |  |    |                |       |   |    |    |        |        |        |        |    |      |                                                             |